Title of the Presentation

Subtitle or Conference Name

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Department / Faculty Name
University / Institution Name

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Outline

Introduction

Motivation

The Problem

- Describe the core problem you are addressing.
- Why is this problem important in AI/ML?
- What are the limitations of current approaches?

Our Contribution

- Briefly state your proposed solution or main findings.
- Highlight key improvements (e.g., accuracy, efficiency).

Background & Related Work

Related Work

- **Approach A [Smith et al., 2021]:** Solves X but struggles with Y.
- **Approach B [Jones et al., 2022]:** Improves Y using technique Z, but is computationally expensive.

Gap in the Literature

There is a need for a model that achieves high accuracy while remaining computationally efficient on edge devices.

Proposed Methodology

Methodology Overview

- Provide a high-level overview of your model architecture or algorithm.
- Use modern diagrams with shadow and clean edges.

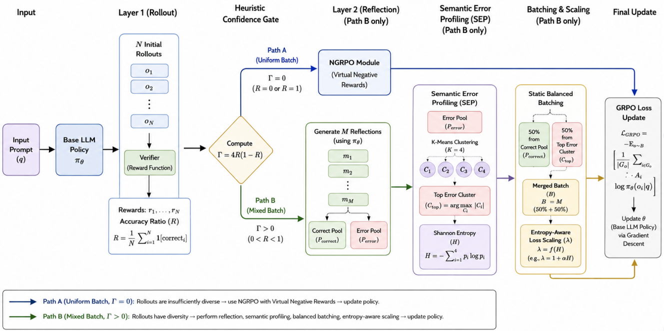

Figure 1: High-level architecture of the proposed approach.

Mathematical Formulation

We optimize the parameters θ by minimizing the empirical risk.

Objective Function

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \ell(f_\theta(\boldsymbol{x}_i), y_i) + \lambda \mathcal{R}(\theta) \tag{1}$$

where:

- $\ell(\cdot, \cdot)$ is the loss function (e.g., Cross-Entropy).
- $\mathcal{R}(\theta)$ is the regularization term to prevent overfitting.
- λ is the regularization hyperparameter.

Experiments & Results

Experimental Setup

Datasets

- ImageNet-1K
- COCO 2017
- Custom Dataset

Hardware & Details

- 4x NVIDIA A100 GPUs
- PyTorch 2.0
- AdamW ($\mathrm{lr} = 10^{-4}$)

Results: Quantitative Comparison

Table 1: Performance comparison on the benchmark dataset.

Method	Accuracy (%)	F1-Score	Params (M)
Baseline 1	85.2	0.84	25.4
Baseline 2	88.7	0.88	42.1
Ours	**91.4**	**0.91**	**18.5**

Our model outperforms all baselines while requiring 27% fewer parameters!

Results: Visualizations

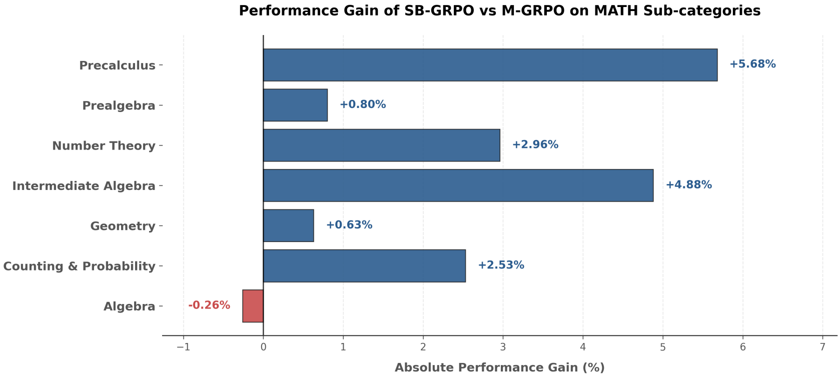

Figure 2: Experimental results showing math delta chart.

Key Takeaway

Our method achieves **state-of-the-art** performance with **27% fewer parameters**, making it practical for real-world edge deployment.

Conclusion

Conclusion & Future Work

Summary

- We proposed a novel, lightweight approach for [Task].
- Achieved state-of-the-art results on [Dataset].
- Reduced computational overhead by 30%, enabling edge deployment.

Future Directions

- Extend the framework to handle multimodal inputs (text + image).
- Investigate theoretical bounds of the optimization landscape.

Thank You!

Questions & Discussion

Appendix

Backup Slides

Ablation Study

Table 2: Ablation on key components.

Configuration	Acc (%)	Δ
Full Model	**91.4**	—
w/o Component A	89.1	-2.3
w/o Component B	87.5	-3.9
w/o Component A & B	85.0	-6.4

Key Findings

- **Component A** gives biggest single impact (-2.3%).
- Synergy of modules yields $+6.4\%$ gain.

Hyperparameter Sensitivity

Key Hyperparameters

Hyperparameter	Search Space	Optimal
Learning Rate	$\{10^{-5}, 10^{-4}, 10^{-3}\}$	10^{-4}
Batch Size	$\{16, 32, 64, 128\}$	64
Weight Decay	$\{0, 10^{-4}, 10^{-2}\}$	10^{-4}
Dropout Rate	$\{0.0, 0.1, 0.3, 0.5\}$	0.1